

# Directional Collapse in Fine-Tuning: Preserving the Way Back

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## Abstract

Forward-only supervised fine-tuning can improve a trained mapping while damaging its backward counterpart, even when the base model already performs both. We study this tradeoff across translation, code transformation, semantic parsing, and formal tasks. Our central result is that a small amount of backward supervision can suffice: replacing just 1% of forward training pairs with their reversals restores formatting backward success from 0.4% to 90.2%, while forward success remains 99.6% versus 99.7% under forward-only SFT in a fixed Llama3B comparison. The intervention uses ordinary cross-entropy, no new paired data, and the same number of training instances and optimizer steps. We compare all dose rungs with generic replay, longer forward training, reverse-only training, contrastive learning, and other auxiliary objectives. Forward-only auxiliary objectives often fail to preserve backward generation; mixed-direction objectives sometimes improve on reversed-pair SFT. Task-dependent costs and unit-conversion null results limit generality. Recovery diagnostics support selective interference in one cell, rather than universal memory erasure. These results make small-dose reversed-pair SFT a necessary baseline before adding objective complexity.

## 1 Introduction

Fine-tuning should improve a task without unnecessarily sacrificing capabilities the model already has. A paired dataset makes this concern concrete: translation can run in either language direction, a code transformation can be undone, and a question–SQL pair supports both parsing and verbalization. Training on one direction can improve that mapping while making the opposite direction fail. A forward-only evaluation would report success and miss the loss.

Prior work establishes directional degradation in code transformation (Nikiema et al., 2025) and

translation (Zhu et al., 2024). We ask a practical question: *if we want to keep the backward task, how much backward training is needed, and does it require a new loss?* Our answer starts with changing the training inputs, rather than the objective. Each example already supplies a backward target; we replace a small fraction of forward pairs with their reversals and train with ordinary cross-entropy.

**The problem and the simple intervention.** Table 1 puts base, forward-only SFT, and every registered adaptation side by side, with both directions expressed as success percentages. In its fixed Llama3B formatting comparison, SFT raises forward success from 60.2% to 99.7% but reduces backward success from 51.1% to 0.4%. Reversing 1% of training pairs yields 99.6% forward and 90.2% backward success. Code transformation shows the same qualitative pattern. These are concrete within-campaign examples, not a claim that every task improves forward or that 1% always suffices.

**Why compare objectives against reversed pairs?** A specialized loss can supply directional information as well as change optimization. Our contrasts separate those contributions using ordinary reversed-pair SFT, generic instruction replay, longer forward training, extra-CE proxies, and shuffled-pair controls. Table 2 shows both directions for contrastive and other auxiliary objectives. Forward-only alignment does not consistently rescue backward generation. Mixed-direction alignment sometimes beats the simple mixture, so our conclusion is that small-dose SFT is an essential baseline, not that auxiliary objectives never help. Replacement matches instances and optimizer steps; realized token exposure and runtime are audited separately.

**What the comparison establishes.** We contribute a task-by-adaptation account of directional loss and preservation, a low-dose ordinary-SFT baseline, and objective comparisons that retain directional and compute controls. We also test whether lost behavior can be recovered. The evidence supports selective interference in a formatting cell but does not identify a universal internal cause. It should not be reduced to memorization or permanent erasure: behavioral failure alone establishes neither. Failed new-domain gates, replicated unit-conversion null results, and recipe-dependent forward costs bound the practical claim. The broader model/seed audit and uncertainty accompany the compact main tables in the appendix and source evidence.

## 2 Directional collapse, defined

A *paired task* is a pair of datasets  $(X, Y)$  and  $(Y, X)$  over the same instances, where the reverse training set is obtained from the forward one by exchanging input and target. We call  $X \rightarrow Y$  the *forward* direction and  $Y \rightarrow X$  the *backward* (or reverse) direction. *Directional collapse* is a fall in reverse-direction success after fine-tuning on forward-direction data alone, measured against the untouched model.

Every domain has a strict per-instance criterion with an automatic check (§3). Two failure modes are tracked separately:

**Echo** the output reproduces its own input. This is not hygiene. In JSON-to-YAML conversion, JSON is valid YAML, so a model that copies its input *parses* and *compares structurally equal* to the target; without the echo guard the task is not a task. Echo rates rose under forward-only training in earlier work, so a criterion that admits them measures the criterion rather than the capability.

**Off-target** the output is in the wrong language, format, or notation.

Reverse strict success excludes both failure modes. The original translation forward predicate combines target-language and quality checks but does not explicitly exclude echoes; its forward-retention interpretation has this limitation. The versioned transfer diagnostic in Amendment51 adds echo exclusion in both directions and retains the original failed gates. Serialization strict scoring excludes echoes in both directions.

Thresholds for continuous criteria are fixed from the *untouched* model’s own score distribution before any fine-tuned model is scored, and are recorded with the date and quantile that produced them.

## 3 Setup

**Domains.** The audited main grid contains 11 task orientations covering translation, code transformation, semantic parsing, factual relations, executable reasoning, format conversion, and algebra. Opposite orientations of the same task are not independent domains. The invented-format never-had control is excluded from this count.

**Models.** The current audited results cover Llama 3B/8B, Gemma 4B/12B, and OLMo 1B. Eligibility and seed coverage vary by task and model; failed gates are retained.

**Arms and budget.** The forward-only baseline (SFT) trains on forward pairs. Each MIX $k$  arm replaces  $k\%$  of those pairs with their own reversal, partitioned by pair identity so that no pair is seen in both directions. Because the reversal *replaces* rather than adds, instance count and optimizer steps are matched to SFT at every dose. We audit realized sequence and supervised-token counts separately: prompt length and unequal side lengths can change token exposure even when the pair count is fixed. [Golovneva et al. \(2024\)](#) distinguish data-matched from compute-matched reverse training; replacement requires no new paired examples. Any compute-saving claim is scoped to the measured training recipe and its audited token exposure.

Generic instruction REPLAY replaces 50% of forward pairs; its matched-dose comparison is with MIX50, while low-dose contrasts also differ in replacement share. REV trains backward only and tests backward learnability, with its forward cost reported. Longer-forward and both-direction controls use additional training budget; all arm definitions and realized direction/token counts remain in the run manifests.

## 4 Forward gains can hide backward losses

Table 1 is the reference comparison: one fixed model and training seed, all eleven audited task orientations, and every registered main-grid adaptation. Paired forward/backward rows make the

| Task                          | Dir. | Reference |             | Reversed-pair share |             |      |      |      | Other adaptations |      |               |      |       |
|-------------------------------|------|-----------|-------------|---------------------|-------------|------|------|------|-------------------|------|---------------|------|-------|
|                               |      | Base      | SFT         | 1%                  | 5%          | 10%  | 25%  | 50%  | Replay            | Rev. | Fwd2 $\times$ | Flip | Multi |
| German $\rightarrow$ English  | F    | 73.7      | 67.1        | <b>67.0</b>         | <b>67.4</b> | 66.2 | 68.6 | 67.4 | 68.6              | 63.5 | 57.8          | 64.4 | 69.1  |
|                               | B    | 79.2      | <b>0.0</b>  | <b>79.4</b>         | <b>82.0</b> | 82.8 | 82.3 | 82.9 | 59.0              | 85.2 | 1.4           | 83.9 | 71.5  |
| English $\rightarrow$ German  | F    | 79.2      | 84.8        | <b>84.6</b>         | <b>85.5</b> | 85.4 | 84.7 | 84.6 | 85.4              | 0.0  | 77.8          | 84.8 | 84.0  |
|                               | B    | 73.3      | 64.4        | <b>70.8</b>         | <b>68.5</b> | 68.2 | 68.2 | 67.8 | 72.0              | 66.2 | 64.4          | 67.0 | 73.0  |
| English $\rightarrow$ Chinese | F    | 74.1      | 77.9        | <b>76.6</b>         | <b>77.4</b> | 77.1 | 78.6 | 76.7 | 78.6              | 0.0  | 75.9          | 80.4 | 77.0  |
|                               | B    | 75.4      | <b>4.0</b>  | <b>68.2</b>         | <b>68.3</b> | 69.9 | 71.5 | 71.9 | 70.9              | 71.2 | 9.5           | 72.8 | 70.3  |
| Chinese $\rightarrow$ English | F    | 76.0      | 71.5        | <b>72.5</b>         | <b>72.1</b> | 72.3 | 72.2 | 71.8 | 74.8              | 23.4 | 63.8          | 71.5 | 73.3  |
|                               | B    | 74.0      | <b>0.0</b>  | <b>76.2</b>         | <b>77.1</b> | 80.0 | 78.3 | 77.2 | 1.0               | 77.7 | 0.0           | 76.1 | 71.0  |
| Code transformation           | F    | 37.5      | 82.4        | <b>82.8</b>         | <b>84.3</b> | 83.6 | 84.4 | 83.9 | 74.8              | 65.6 | 88.3          | 86.3 | 70.5  |
|                               | B    | 32.0      | <b>3.8</b>  | <b>55.8</b>         | <b>75.2</b> | 76.5 | 80.3 | 85.6 | 14.5              | 86.3 | 9.6           | 86.6 | 3.4   |
| Question $\rightarrow$ SQL    | F    | 49.5      | 62.4        | <b>62.3</b>         | <b>60.8</b> | 62.0 | 60.1 | 58.6 | 59.8              | 42.0 | 61.1          | 60.8 | 56.8  |
|                               | B    | 51.1      | <b>23.2</b> | <b>48.5</b>         | <b>55.0</b> | 57.5 | 59.9 | 58.1 | 17.1              | 59.1 | 27.3          | 58.8 | 50.4  |
| Format conversion             | F    | 60.2      | 99.7        | <b>99.6</b>         | <b>99.9</b> | 99.8 | 99.8 | 99.8 | 99.0              | 25.2 | 99.0          | 99.8 | 97.8  |
|                               | B    | 51.1      | <b>0.4</b>  | <b>90.2</b>         | <b>96.2</b> | 97.0 | 98.6 | 99.6 | 24.6              | 99.4 | 0.7           | 99.6 | 42.9  |
| Factual relations             | F    | 71.9      | 86.7        | –                   | –           | –    | 86.7 | 87.4 | 88.1              | 79.3 | 89.6          | 89.6 | 83.0  |
|                               | B    | 65.2      | 71.1        | –                   | –           | –    | 80.0 | 80.7 | 67.4              | 83.7 | 58.5          | 81.5 | 65.9  |
| Expand $\rightarrow$ factor   | F    | 17.3      | 26.1        | <b>25.5</b>         | <b>26.8</b> | 27.3 | 25.5 | 24.8 | 25.8              | 14.6 | 29.6          | 27.7 | 21.2  |
|                               | B    | 14.7      | 7.8         | <b>14.6</b>         | <b>20.6</b> | 25.3 | 31.0 | 36.5 | 9.2               | 43.6 | 6.7           | 44.9 | 6.0   |
| Factor $\rightarrow$ expand   | F    | 14.6      | 43.6        | <b>44.0</b>         | <b>43.8</b> | 43.2 | 40.2 | 36.1 | 36.9              | 7.8  | 49.6          | 44.7 | 26.8  |
|                               | B    | 17.4      | 14.7        | <b>10.6</b>         | <b>15.8</b> | 18.0 | 23.0 | 24.9 | 14.0              | 26.2 | 12.7          | 27.4 | 26.5  |
| Execution prediction          | F    | 26.5      | 29.0        | –                   | –           | –    | 29.5 | 29.5 | 27.0              | 21.0 | 31.5          | 30.5 | 24.5  |
|                               | B    | 16.5      | 13.0        | –                   | –           | –    | 26.0 | 28.0 | 13.0              | 30.0 | 12.5          | 29.0 | 12.5  |

Table 1: Strict success (%); F/B denotes forward/backward. Blue highlights small reversed-pair doses; red highlights SFT backward success at most half the same-pass base (a descriptive threshold). Llama 3.2-3B, fixed seed 17, all eleven audited task orientations. Dose columns replace that share of forward pairs with their reversals, keeping instances and steps fixed. Replay replaces 50% with generic instructions; Rev. trains backward only; Fwd2 $\times$  doubles forward training; Flip trains both directions at doubled budget; Multi mixes other tasks. Dashes are unresolved dose rungs, not zeros.

tradeoff visible without pooling tasks or choosing successful seeds. Red marks the registered descriptive collapse threshold; blue identifies low-dose mixtures, including their adverse results. Unresolved dose rungs are explicit.

The broader snapshot contains 91 distinct audited model–task–seed cells from 103 validated campaigns. Forward SFT leaves backward success at most half the same-pass base rate in 52 cells. Repeated passes are selected by latest valid completion time, independently of effect sign. These are descriptive counts, not independent-domain counts or pooled significance tests (Appendix A). Forward gains are not universal: translation orientations include forward degradation, while factual relations do not collapse in the reference comparison.

General-ability probes separately measure instruction following and reasoning, with overlap exclusions recorded before generation (Appendix B.2). Their endpoint-specific changes preclude interpreting task scores alone as disproportionate forgetting or universal preservation.

## 5 Small backward doses with ordinary SFT

The simplest preservation baseline changes only which side of an existing pair is the input. In the reference formatting cell, 1% reversal raises backward success from 0.4% to 90.2%; 5% reaches 96.2%. Forward success stays close to forward-only SFT. For code transformation, 1% raises backward success from 3.8% to 55.8% while forward success changes from 82.4% to 82.8%. Generic replay and doubled forward training do not provide comparable backward recovery in these same passes (Table 1). The contrast is particularly useful: more forward optimization is not a substitute for correct backward targets.

The full dose ladder also shows why these examples cannot justify a universal 1% rule. Some tasks need higher doses; algebra includes an adverse low-dose result. We define the registered knee as the smallest dose reaching 90% of MIX50’s backward success, and report paired uncertainty and seed variation rather than treating adjacent rungs as equivalent. Figure 1 retains the units null along-

side collapse cells. Forward retention, general-ability endpoints, measured runtime and token exposure are separate quantities; Appendix B.1 reports their joint audit.

## 6 Boundary conditions: information and inverse difficulty

The format ladder varies whether an inverse is determined, while algebra orientation and automata probe inverse search difficulty. Several ladder cells fail eligibility and cannot support tuned-model comparisons; their descriptive base results remain separate.

**Two kinds of hardness.** Information loss can prevent a unique inverse even with unlimited search, whereas a determined inverse can remain computationally difficult. Failed gates do not distinguish these explanations by themselves.

## 7 Is the reverse capability still accessible?

Four instruments, reported as separate verdicts rather than averaged: they measure different things, and a disagreement between them is a finding rather than noise.

**Elicitation.** The audited reports include both prompt recovery and little recovery under tested prompts. Best-prompt summaries are exploratory; behavioral recovery does not prove a specific internal mechanism.

**Adapter scaling.** Scaling curves retain forward and reverse success together. Forward-progress ratios are undefined when the full update has no positive forward gain, and a curve without reverse loss cannot locate a collapse threshold.

**Relearning against a never-had control.** Relearning curves are matched to the never-had control by model and seed. A small-sample recovery advantage is descriptive evidence of accessible capability; its uncertainty and starting loss must accompany the comparison. This is also where the contrast with the reversal curse (Berglund et al., 2024) becomes experimental rather than rhetorical: that literature describes capabilities never acquired, and our never-had control is that case, constructed.

**Update repair.** LoRA spectral sweeps are exploratory median-relative filters on an already low-

rank update, with no full-rank noise bulk. Selecting a best threshold on test success is exploratory. Full-FT projection repair uses a separately registered fixed threshold and has produced 1 checkpoint passing saved-weight audits. The completed fresh generation campaign improves forward and reverse success relative to the original full-SFT checkpoint. This single exploratory repair does not establish a general remedy.

**Explanatory hypothesis and tests.** Five completed seed-17 diagnostics test task-conditioned output selection (Appendix C). In formatting and translation collapse cells, forward SFT reduces reverse gold-over-copy preference and mixed-direction SFT restores it. Absolute reverse-answer loss worsens in Llama but improves in Gemma translation despite collapsed generation: answer likelihood and generation are distinct. Final-checkpoint gradient cosines provide little support for persistent gradient conflict. Local reverse updates improve held-out loss even in the units null cell, so improvement alone does not diagnose collapse. Formatting middle-band removals recover reverse success beyond scaling and random removals matched for retained update norm; all bands and both directions remain reported. Edit norms are not also matched. These exploratory findings support selective interference in one cell, without establishing stored knowledge or a universal mechanism. Direction probes still require input-cue controls; the never-had task differs from the recovered tasks, and LoRA preserves base weights by construction. Gemma diagnostics exclude inactive vision factors. The remaining registered diagnostics are running in the background; their pending status supplies no evidence.

## 8 Consequences for attribution

We distinguish exposure to both paired sides from an objective-specific generation benefit.

A gain from an auxiliary objective can reflect its directional information, its additional computation, or the objective itself. Our paired contrastive arms combine forward-only or mixed-direction CE with symmetric representation alignment. Both sides are encoded independently, and negatives are drawn only from training data. The contrastive term has no reverse generation CE, but it still exposes inverse correspondence; it cannot be described as having no inverse supervision.

Table 2 reports fresh, complete passes for

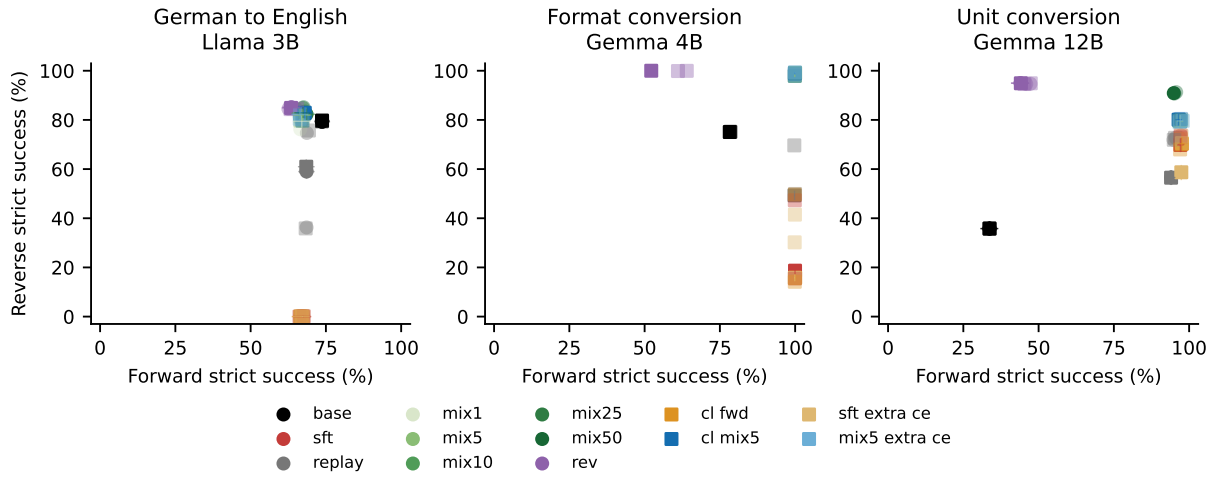


Figure 1: Current preservation frontiers on the fixed reporting panel. Circles show ordinary SFT/dose campaigns; squares show separately generated contrastive/control campaigns. Opaque points and marginal cluster-bootstrap intervals are seed-17 results; translucent points show other training seeds. Contrasts remain paired within each campaign and are retained with measured runtime/exposure records in the source-backed analysis. The units null-collapse boundary case is included. No cross-campaign pooled test is implied.

the registered contrastive comparison and auxiliary objectives that pass the current exposure/withdrawal audit. Each row pair has its own base; numerical differences across evaluation campaigns are not treatment effects. The contrastive grid currently completes 24 of 27 registered model–task–seed cells. Appendix A retains every registered primary contrast, per-seed uncertainty and exposure control.

Forward-only contrastive alignment provides no consistent benefit over extra-CE or shuffled controls. In the reference code cell, conflict-aware fine-tuning falls well short of 5% reversed-pair SFT on backward generation; forward-only unlikelihood also fails to preserve backward translation in the audited cells. This supports trying backward examples before adding an auxiliary objective. It does not establish that every loss baseline is inferior: mixed-direction alignment sometimes beats 5% SFT, and round-trip training wins backward success in one SQL cell while trading forward performance. Extra-CE controls match an exposure proxy, not identical measured FLOPs. An interval spanning zero does not establish equivalence, and degenerate intervals at a generation floor require particular caution.

**New-domain eligibility.** The 6 original small-model task–model gates failed the unchanged eligibility criteria (Table 3). Complete test-set gates and scorer audits are retained. No tuned pilot is admitted for these failed cells. Explicit-output

variants reuse their corpora; they are prompt diagnostics and cannot be counted as independent new-domain successes. The 6 completed explicit-format gates also fail the unchanged criteria. Larger-model feasibility gates on the original tasks are separately registered. Gemma12B passes the unit-conversion gate and completes valid replicated campaigns (Table 4). Forward-only SFT improves both directions; this is retained as a replicated null-collapse result. Replication seeds are admitted by validity, independently of effect sign. Llama8B also passes Python–C++ translation after a technical parser retry; its pilot is evaluated under the registered same-pass protocol. Boolean gates remain failed, with correctness criteria unchanged.

**Full-weight checks.** Full-FT training is complete; 6 of 6 validated evaluation campaigns are complete in the current snapshot. Full checkpoints require isolated engines, unlike the single-engine multi-LoRA comparisons. Their newly generated campaigns use the same instances, messages, tokenizer, and scoring settings but span engine restarts. We record this layout and interpret small differences against the decoding floor. Projection-matrix spectral repair is exploratory; it does not assume a validated noise model. Repair results require joint reverse/forward interpretation.

| (a) Contrastive objectives and exposure controls |       |      |      |             |             |        |       |       |      |         |       |       |
|--------------------------------------------------|-------|------|------|-------------|-------------|--------|-------|-------|------|---------|-------|-------|
| Task                                             | Model | Dir. | Base | SFT         | 5%          | Replay | Rev.  | CL-F  | CL-M | Shuffle | CE-F+ | CE-M+ |
| German → English                                 | G12B  | F    | 76.9 | 66.9        | <b>66.5</b> | 67.3   | 1.6   | 66.1  | 66.8 | 66.1    | 66.5  | 66.6  |
|                                                  |       | B    | 72.0 | <b>0.2</b>  | <b>60.1</b> | 56.2   | 66.4  | 0.2   | 60.9 | 0.3     | 0.1   | 60.4  |
| German → English                                 | G4B   | F    | 75.4 | 67.7        | <b>67.4</b> | 68.7   | 40.0  | 67.2  | 69.4 | 67.1    | 66.9  | 66.7  |
|                                                  |       | B    | 75.8 | <b>0.0</b>  | <b>74.4</b> | 70.4   | 76.8  | 0.0   | 74.5 | 0.0     | 0.0   | 75.1  |
| German → English                                 | L8B   | F    | 74.5 | 64.2        | <b>65.8</b> | 63.9   | 4.7   | 62.8  | 64.5 | 63.8    | 62.8  | 65.1  |
|                                                  |       | B    | 73.0 | <b>16.5</b> | <b>68.9</b> | 64.3   | 75.1  | 0.0   | 67.7 | 0.0     | 18.2  | 69.6  |
| German → English                                 | L3B   | F    | 73.7 | 67.3        | <b>67.0</b> | 68.5   | 63.4  | 66.5  | 68.0 | 66.7    | 67.0  | 66.4  |
|                                                  |       | B    | 79.7 | <b>0.0</b>  | <b>82.0</b> | 61.0   | 84.9  | 0.0   | 82.9 | 0.1     | 0.0   | 82.2  |
| Format conversion                                | G4B   | F    | 78.3 | 100.0       | <b>99.9</b> | 99.8   | 52.2  | 100.0 | 99.9 | 99.9    | 99.9  | 100.0 |
|                                                  |       | B    | 75.1 | <b>18.7</b> | <b>98.0</b> | 49.3   | 100.0 | 15.8  | 99.0 | 45.0    | 15.8  | 99.1  |
| Format conversion                                | L8B   | F    | 80.2 | 99.5        | <b>99.9</b> | 99.8   | 2.2   | 100.0 | 99.8 | 99.7    | 100.0 | 99.9  |
|                                                  |       | B    | 69.0 | <b>0.0</b>  | <b>98.7</b> | 32.1   | 99.9  | 2.2   | 99.2 | 0.0     | 0.0   | 98.7  |
| Format conversion                                | L3B   | F    | 60.6 | 99.7        | <b>99.9</b> | 98.9   | 27.0  | 99.6  | 99.6 | 99.7    | 99.2  | 99.7  |
|                                                  |       | B    | 51.4 | <b>0.4</b>  | <b>96.0</b> | 25.6   | 99.4  | 0.1   | 96.0 | 0.8     | 0.6   | 94.2  |
| Unit conversion                                  | G12B  | F    | 33.7 | 97.1        | <b>97.0</b> | 94.0   | 44.1  | 97.6  | 96.5 | 97.5    | 97.4  | 97.1  |
|                                                  |       | B    | 35.8 | 69.8        | <b>79.4</b> | 56.5   | 94.9  | 70.3  | 80.1 | 74.1    | 58.7  | 79.6  |

| (b) Other auxiliary objectives |       |      |      |             |             |      |              |            |  |
|--------------------------------|-------|------|------|-------------|-------------|------|--------------|------------|--|
| Task                           | Model | Dir. | Base | SFT         | 5%          | CFT  | Unlikelihood | Round-trip |  |
| English → German               | L8B   | F    | 71.1 | 73.1        | <b>73.0</b> | —    | 74.0         | —          |  |
|                                |       | B    | 74.2 | 58.7        | <b>69.2</b> | —    | 0.1          | —          |  |
| English → German               | L3B   | F    | 79.7 | 84.8        | <b>85.4</b> | —    | 85.1         | —          |  |
|                                |       | B    | 72.8 | 63.6        | <b>68.8</b> | —    | 0.0          | —          |  |
| Code transformation            | L3B   | F    | 38.0 | 81.4        | <b>84.2</b> | 84.1 | —            | —          |  |
|                                |       | B    | 32.0 | <b>3.6</b>  | <b>75.4</b> | 6.7  | —            | —          |  |
| Question → SQL                 | L8B   | F    | 62.0 | 72.1        | <b>71.9</b> | —    | —            | 71.3       |  |
|                                |       | B    | 54.9 | 51.3        | <b>57.8</b> | —    | —            | 58.8       |  |
| Question → SQL                 | L3B   | F    | 49.1 | 62.3        | <b>61.0</b> | —    | —            | 61.5       |  |
|                                |       | B    | 51.5 | <b>23.3</b> | <b>55.2</b> | —    | —            | 48.2       |  |

Table 2: Strict success (%); F/B denotes forward/backward. Blue highlights small reversed-pair doses; red highlights SFT backward success at most half the same-pass base (a descriptive threshold). All eligible original contrastive and audited auxiliary-objective cells at fixed seed 17. Each task/model row pair uses its own fresh evaluation pass and base; scores across campaigns are not treatment contrasts. L/G denotes Llama/Gemma; B denotes billions of parameters. The 5% column is ordinary reversed-pair SFT. CL-F/M adds alignment to forward/mixed-direction CE; Shuffle randomizes positive pairs; CE-F+/M+ adds CE as an exposure proxy; CFT is conflict-aware fine-tuning. Dashes denote untested objectives. Failed gates, other seeds and paired intervals remain in the appendix/audit.

## 9 Related work

Prior directional degradation and reverse training are established starting points. Our contribution is the joint preservation comparison and controlled attribution, qualified by failed gates and replicated null-collapse results.

**Directional degradation and reverse training.** Nikiema et al. (2025) study cognitive specialization in code transformation; Zhu et al. (2024) examine direction-dependent degradation in translation. We ask how preservation varies across domains, doses, models, and objectives. Program-execution reasoning also exhibits directional asymmetry (Hasanov et al., 2026), motivating cell-specific measurement of starting reverse

capability. The reversal curse concerns capabilities never acquired (Berglund et al., 2024); reverse training addresses this at the data level under data- and compute-matched comparisons (Golovneva et al., 2024). Our relearning experiment compares a collapsed model with a model-matched never-had control. Replacement mixtures hold instances and steps fixed while auditing realized exposure.

**Forgetting and rehearsal.** Broader forgetting after fine-tuning (Luo et al., 2023) and differences between LoRA and full fine-tuning (Biderman et al., 2024) motivate our general-ability and full-weight controls; they do not bound directional loss in these tasks. Rehearsal can preserve earlier performance (Scialom et al., 2022). Generic replay at

| Task   | Model      | Forward | Reverse | Eligibility |
|--------|------------|---------|---------|-------------|
| units  | llama32-3b | 0.4     | 1.9     | Fail        |
| units  | gemma3-4b  | 37.6    | 3.5     | Fail        |
| logic  | llama32-3b | 0.0     | 0.0     | Fail        |
| logic  | gemma3-4b  | 0.0     | 0.0     | Fail        |
| py_cpp | llama32-3b | 78.6    | 74.1    | Fail        |
| py_cpp | gemma3-4b  | 5.1     | 99.9    | Fail        |

Table 3: Original new-domain base gates: strict success in percent. Failed cells are retained and receive no tuned pilot. Explicit-output diagnostics use the same semantic splits and are separate exploratory variants.

the registered replacement share tests whether directional content adds preservation beyond that explanation. Concentration of inverse loss relative to general ability remains a separate empirical question.

**Recovery and task inference.** Wrapper and off-set accounts of fine-tuning (Jain et al., 2024; Lee et al., 2024), update filtering (spe, 2026), and skewed task inference (Kotha et al., 2024) motivate distinct recovery instruments. Our directional tests include a never-had control, both-direction retention, and fixed-threshold exploratory repair. Behavioral recovery does not identify a unique internal mechanism, and bidirectional behavior need not imply a unified representation (lat, 2026).

**Attributing objective gains.** Data mixtures and reporting choices can confound objective attribution (Longpre et al., 2023; Lipton and Steinhardt, 2018). RevThink adds backward questions and reasoning (Chen et al., 2025); its cross-entropy components illustrate why directional exposure must be controlled when assessing a bidirectional objective. Our paired contrastive comparison includes reversed-pair SFT, extra-CE proxies, and shuffled positives, measuring generation in both directions. These controls test the scope of an objective-specific gain without assuming that auxiliary objectives are universally unnecessary.

## 10 Conclusion

Forward-only SFT can improve the trained task while damaging an existing backward task. Small-dose reversed-pair SFT directly addresses that failure with ordinary cross-entropy: 1% reversal gives substantial backward recovery with little forward change in the reference formatting and code cells. The complete tables retain harder tradeoffs, nulls, and objective-specific gains. Evaluate both directions and establish this simple baseline before attributing preservation to a more complex loss. Recovery evidence motivates selective interference

as a hypothesis, without proving memory erasure or a universal cause.

## Limitations

**Where the repair does not reach.** Reversed-pair training cannot recover information absent from an input. Our failed new-domain gates also limit generality beyond established tasks.

**Scale.** The full arm set runs at 1–4B with three seeds; only core arms run at 8–12B with two where eligible in the original grid. Separately registered larger-model extensions add replication and objective controls in selected cells; coverage remains uneven. This does not establish scaling to larger models.

**LoRA.** Most arms use LoRA. Prior work reports different retention under full fine-tuning (Biderman et al., 2024); this does not guarantee a conservative directional-loss estimate in our tasks. The full-FT check covers only the registered Llama 3B seed-17 cells; it cannot bound directional loss across other models or training recipes.

**Resolution of the dose ladder.** Adjacent rungs differ by a couple of points, which our evaluation sets may not separate reliably. Final dose claims require intervals and seed ranges; an interval spanning zero does not establish equivalence.

**Criteria with a model in them.** Two directions are scored by round-tripping through a frozen model rather than by execution. Each reports that model’s own accuracy on the references as the criterion’s ceiling where measured. Extractor/parser errors can reject valid generations; the D2T measurement is still undergoing validation.

**Languages and domains.** The original code-transformation cell is Python only; the translation cells cover two language pairs, both with English

| Task   | Model      | Seed | Arm    | Forward | Reverse |
|--------|------------|------|--------|---------|---------|
| py_cpp | llama31-8b | 17   | base   | 85.9    | 100.0   |
| py_cpp | llama31-8b | 17   | sft    | 100.0   | 0.0     |
| py_cpp | llama31-8b | 17   | mix50  | 100.0   | 48.7    |
| py_cpp | llama31-8b | 17   | replay | 100.0   | 97.6    |
| py_cpp | llama31-8b | 17   | rev    | 0.0     | 0.0     |
| py_cpp | llama31-8b | 42   | base   | 84.2    | 100.0   |
| py_cpp | llama31-8b | 42   | sft    | 100.0   | 0.0     |
| py_cpp | llama31-8b | 42   | mix50  | 1.7     | 0.0     |
| py_cpp | llama31-8b | 42   | replay | 99.9    | 99.5    |
| py_cpp | llama31-8b | 42   | rev    | 0.0     | 0.0     |
| units  | gemma3-12b | 1234 | base   | 33.7    | 35.8    |
| units  | gemma3-12b | 1234 | sft    | 96.9    | 72.7    |
| units  | gemma3-12b | 1234 | mix50  | 95.1    | 90.7    |
| units  | gemma3-12b | 1234 | replay | 94.8    | 71.9    |
| units  | gemma3-12b | 1234 | rev    | 47.3    | 94.7    |
| units  | gemma3-12b | 17   | base   | 33.7    | 35.8    |
| units  | gemma3-12b | 17   | sft    | 97.4    | 70.0    |
| units  | gemma3-12b | 17   | mix50  | 94.9    | 90.9    |
| units  | gemma3-12b | 17   | replay | 94.0    | 56.6    |
| units  | gemma3-12b | 17   | rev    | 44.0    | 95.0    |
| units  | gemma3-12b | 42   | base   | 33.7    | 35.8    |
| units  | gemma3-12b | 42   | sft    | 97.3    | 73.7    |
| units  | gemma3-12b | 42   | mix50  | 95.7    | 91.3    |
| units  | gemma3-12b | 42   | replay | 95.5    | 72.7    |
| units  | gemma3-12b | 42   | rev    | 45.9    | 94.6    |
| py_cpp | llama31-8b | 1234 | base   | 84.2    | 100.0   |
| py_cpp | llama31-8b | 1234 | sft    | 100.0   | 0.0     |
| py_cpp | llama31-8b | 1234 | mix50  | 100.0   | 0.2     |
| py_cpp | llama31-8b | 1234 | replay | 95.8    | 97.9    |
| py_cpp | llama31-8b | 1234 | rev    | 0.0     | 0.0     |

Table 4: Eligible larger-model generality campaigns: strict success in percent, with all arms and the base evaluated in one fresh pass per model/task/seed. The completed unit-conversion replications improve both directions under SFT and are retained null-collapse boundary cases. Additional eligible campaigns enter this table after validated analysis.

on one side. Synthetic task variants do not establish natural code or unrestricted reasoning generality, and reused semantic corpora are not independent evidence. Independent transfer retains failed eligibility boundaries; completed prompt and training-recipe tests retain adverse and null findings (Appendices B.3 and B.4).

## References

2026. The illusion of latent generalization: Birectionality and the reversal curse. *Preprint*, arXiv:2604.04943. Probes are consistent with forward and reverse stored as distinct entries rather than one direction-agnostic representation.
2026. Spectral unforgetting: Post-hoc recovery of damaged capabilities without retraining. *Preprint*, arXiv:2605.20296.
- Lukas Berglund, Meg Tong, Max Kaufmann, Mikita Balesni, Asa Cooper Stickland, Tomasz Korbak, and Owain Evans. 2024. The reversal curse: Lms trained on “a is b” fail to learn “b is a”. In *International Conference on Learning Representations*.

- Dan Biderman, Jacob Portes, Jose Javier Gonzalez Ortiz, et al. 2024. Lora learns less and forgets less. *Transactions on Machine Learning Research*. Read 2026-09-11. LoRA underperforms full fine-tuning on target domains but better maintains out-of-domain performance. Qualifies every LoRA-based claim here.
- Justin Chih-Yao Chen, Zifeng Wang, Hamid Palangi, Rujun Han, Sayna Ebrahimi, Long T. Le, Vincent Perot, Swaroop Mishra, Mohit Bansal, Chen-Yu Lee, and Tomas Pfister. 2025. Reverse thinking makes lms stronger reasoners. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics*. Measures FORWARD accuracy: backward data added to improve forward reasoning. Loss is a sum of three cross-entropy terms over a union of formats (p. 4), and no baseline holds data direction fixed.
- Olga Golovneva, Zeyuan Allen-Zhu, Jason Weston, and Sainbayar Sukhbaatar. 2024. Reverse training to nurse the reversal curse. *Preprint*, arXiv:2403.13799.
- Eshgin Hasanov, Md. Mahadi Hassan Sibat, Shubhra Kanti Karmaker, and Aashish Yadavally. 2026. The path not taken: Duality in reasoning

| Task     | Seed | Direction | Base | Full SFT | Full mix5 |
|----------|------|-----------|------|----------|-----------|
| code     | 17   | forward   | 37.9 | 70.0     | 70.7      |
| code     | 17   | reverse   | 31.9 | 2.0      | 65.9      |
| mt_de-en | 17   | forward   | 73.3 | 70.9     | 72.0      |
| mt_de-en | 17   | reverse   | 80.1 | 72.8     | 83.6      |
| mt_en-de | 17   | forward   | 80.1 | 87.0     | 86.6      |
| mt_en-de | 17   | reverse   | 72.9 | 69.9     | 72.4      |
| mt_en-zh | 17   | forward   | 73.9 | 79.1     | 79.2      |
| mt_en-zh | 17   | reverse   | 75.9 | 66.9     | 71.8      |
| mt_zh-en | 17   | forward   | 76.3 | 74.2     | 73.9      |
| mt_zh-en | 17   | reverse   | 73.8 | 70.2     | 78.5      |
| sql      | 17   | forward   | 49.3 | 60.5     | 59.7      |
| sql      | 17   | reverse   | 50.4 | 18.7     | 55.1      |

Table 5: Completed Llama3B full-weight campaign rates in percent, with forward and reverse retention. Checkpoints use isolated engines and identical held-out instances; these comparisons span engine restarts. Paired intervals are recorded in followup contrasts. An empty body means no validated full campaign has completed.

about program execution. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*. DexBench: 445 paired instances, 13 LLMs evaluated. No fine-tuning. Code and data at <https://github.com/sail-ucf/dexbench>.

Samyak Jain, Robert Kirk, Ekdeep Singh Lubana, et al. 2024. Mechanistically analyzing the effects of fine-tuning on procedurally defined tasks. In *International Conference on Learning Representations*. Read 2026-09-11. Fine-tuning rarely alters underlying capabilities; a minimal ‘wrapper’ is learned on top.

Suhas Kotha, Jacob Mitchell Springer, and Aditi Raghunathan. 2024. Understanding catastrophic forgetting in language models via implicit inference. In *International Conference on Learning Representations*. Read 2026-09-11. Fine-tuning skews IMPLICIT TASK INFERENCE; conjugate prompting recovers pretraining capability. A precedent for H1, not merely for forgetting.

Andrew Lee, Xiaoyan Bai, Itamar Pres, et al. 2024. A mechanistic understanding of alignment algorithms: A case study on dpo and toxicity. In *International Conference on Machine Learning*. Read 2026-09-11. Pretraining capabilities are NOT removed but BYPASSED, via an offset distributed amongst layers. Bears on the layer-ablation experiment.

Zachary C. Lipton and Jacob Steinhardt. 2018. Troubling trends in machine learning scholarship. *arXiv preprint arXiv:1807.03341*. Read 2026-09-11. Names ‘failure to identify the sources of empirical gains’ as a trend; our attribution section is an instance with direction in place of hyper-parameters.

Shayne Longpre, Le Hou, Tu Vu, et al. 2023. The flan collection: Designing data and methods for effective instruction tuning. In *International Conference on Machine Learning*. Read 2026-09-11. Task balancing and enrichment are critical and overlooked; gains attributed to method often belong to the mixture.

Yun Luo, Zhen Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. 2023. *An empirical study of catastrophic forgetting in large language models during continual fine-tuning*. Preprint, arXiv:2308.08747. Read 2026-09-11. Forgetting across 1b-7b during continual instruction tuning, worsening with scale in that range.

Serge Lionel Nikiema, Jordan Samhi, Micheline Moumoula, Albarick Djié, Abdoul Kader Kaboré, Jacques Klein, and Tegawendé F. Bissyandé. 2025. *Using contrastive learning to improve two-way reasoning in large language models: The obfuscation task as a case study*. Names “cognitive specialization”: forward-only fine-tuning degrades pre-existing bidirectional reasoning. Declares a Bidirectional Fine-Tuning baseline (p. 6) and reports no number for it.

Thomas Scialom, Tuhin Chakrabarty, and Smaranda Muresan. 2022. Fine-tuned language models are continual learners. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Read 2026-09-11. 1 percent memory buffer preserves 100 percent of prior performance across 8 sequential tasks. The buffer replays ACTUAL earlier-task examples and ADDS to the budget; our dose is the same pairs reversed and REPLACES. Our replay arm is their rehearsal, and the prediction is that it does NOT cure directional collapse.

Dawei Zhu, Pinzhen Chen, Miaoran Zhang, Barry Haddow, Xiaoyu Shen, and Dietrich Klakow. 2024. *Fine-tuning large language models to translate: Will a touch of noisy data in misaligned languages suffice?* In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*. Single-direction fine-tuning generalizes, EXCEPT that English on the target side causes task misinterpretation damaging translation into non-English.

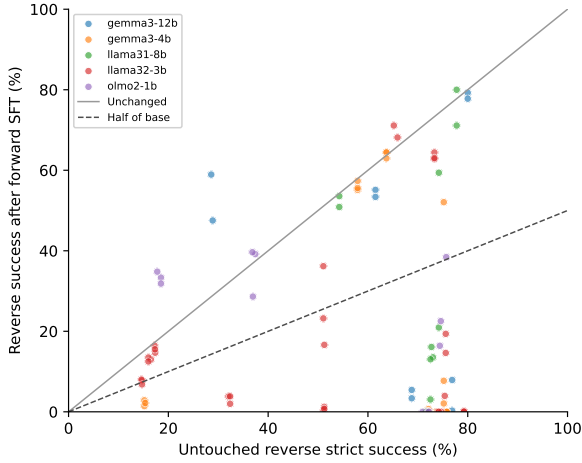


Figure 2: Audited model–task–seed cells, each using a same-pass base and SFT. The dashed line marks half the base rate. Repeated tasks and seeds are not independent observations.

## A Complete comparison counts and objective contrasts

The main tables show strict success percentages for a fixed reference seed. The following inventory covers the broader audited grid; contrastive tables retain all registered paired contrasts with their uncertainty and controls. Highlighting in the main tables identifies arms and the descriptive collapse threshold, not statistical significance.

## B Paper-completion protocol and reporting plan

This appendix records frozen exploratory additions registered before their measurements. Amendment54 freezes new training and replication for a 24-hour paper sprint. Allocation-local smokes precede automatic admission of each new production workload. Registration and queue admission do not establish completed results. The sprint retains completed evidence and five fixed diagnostic jobs. Failed gates and the replicated Gemma12B unit-conversion null-collapse result remain visible.

### B.1 Dose, forward retention, and computational cost

The main figure plots forward and reverse strict success for the base model, forward SFT, generic replay, available reversed-pair doses, and contrastive/control arms. Panels retain model, task orientation, and seed identity. Arms lacking a common evaluation campaign are not connected

as paired observations. The units boundary case is shown alongside collapse cells. Repeated passes use the latest validated completion time, with earlier passes retained in the audit; selection never uses effect sign.

Paired contrasts use the domain’s instance/cluster unit. Training-seed variation is separate: resampling test instances does not measure training variability. No aggregate pools cells as independent domains. The dose knee uses the criterion in §5; overlapping intervals do not establish equivalence.

The cost table reports available trainer runtime, optimizer steps, and CE-corpus tokens. Auxiliary encoded-side exposure is retained separately in the analysis. Missing timing, GPU-type, queue-time, or exposure records remain unavailable. Additional CE is the declared proxy; corpus token counts alone do not establish matched objective FLOPs or wall time.

**Scoped finding.** On the representative translation campaign, reversed-pair SFT preserves reverse behavior with nearly unchanged forward task performance relative to forward SFT (Table 10). Generic replay recovers less reverse performance in that campaign. The units case improves both ways under ordinary SFT; robustness results show that preservation and forward costs depend on task and recipe. Recorded budgets support within-recipe comparisons, not matched wall-time claims across different objectives.

### B.2 General ability and overlap controls

The fixed panel is German-to-English/Llama3B, format conversion/Gemma4B, and units/Gemma12B. Each campaign compares base, SFT, replay, mixed-direction SFT, forward-only and mixed-direction contrastive arms, and their additional-CE controls. The auxiliary arm is fixed rather than selected by its best test score.

IFEval measures instruction following without demonstrations; GSM8K measures numerical reasoning using fixed training demonstrations. A neutral assistant system prompt is shared by all systems. Dataset revisions and overlap exclusions are recorded before generation. Auditing covers both sides of every local training/validation corpus, replay, and mixed-task sources, using normalized exact/contained strings. Fixed demonstrations are checked against the retained test set. Lexical auditing does not establish semantic or pretraining inde-

| Model      | Task orientations | Cells | Threshold met |
|------------|-------------------|-------|---------------|
| gemma3-12b | 7                 | 14    | 8             |
| gemma3-4b  | 7                 | 21    | 14            |
| llama31-8b | 6                 | 12    | 7             |
| llama32-3b | 11                | 32    | 18            |
| olmo2-1b   | 4                 | 12    | 5             |

Table 6: Audited model–task–seed cells. Threshold met means SFT reverse strict success is at most half the same-pass base rate. Counts are descriptive and are not independent task counts or pooled significance tests.

| Task     | Model      | Seed | $\Delta$ CL-fwd versus SFT | $\Delta$ CL-mix5 versus mix5 |
|----------|------------|------|----------------------------|------------------------------|
| fnt      | gemma3-4b  | 17   | -2.88 [-3.88, -1.96]       | +1.08 [+0.32, +1.88]         |
| fnt      | gemma3-4b  | 42   | -1.28 [-1.96, -0.64]       | -1.04 [-1.80, -0.32]         |
| fnt      | gemma3-4b  | 1234 | +2.36 [+1.60, +3.20]       | -0.32 [-0.92, +0.24]         |
| fnt      | llama31-8b | 17   | +2.24 [+1.52, +3.04]       | +0.48 [-0.12, +1.12]         |
| fnt      | llama31-8b | 42   | -33.12 [-35.48, -30.64]    | +0.40 [-0.20, +1.00]         |
| fnt      | llama31-8b | 1234 | +0.00 [+0.00, +0.00]       | -0.72 [-1.52, +0.12]         |
| fnt      | llama32-3b | 17   | -0.24 [-0.56, +0.04]       | +0.00 [-1.12, +1.12]         |
| fnt      | llama32-3b | 42   | -1.48 [-2.12, -0.92]       | +1.68 [+0.52, +2.88]         |
| fnt      | llama32-3b | 1234 | -0.52 [-1.00, -0.04]       | +4.44 [+3.08, +5.84]         |
| mt_de-en | gemma3-12b | 17   | +0.00 [+0.00, +0.00]       | +0.79 [-1.78, +3.56]         |
| mt_de-en | gemma3-12b | 42   | -0.10 [-0.40, +0.00]       | +0.59 [-2.08, +3.26]         |
| mt_de-en | gemma3-12b | 1234 | +0.00 [+0.00, +0.00]       | +0.69 [-1.68, +3.16]         |
| mt_de-en | gemma3-4b  | 17   | +0.00 [+0.00, +0.00]       | +0.10 [-2.27, +2.27]         |
| mt_de-en | gemma3-4b  | 42   | +0.00 [+0.00, +0.00]       | +0.40 [-2.08, +2.77]         |
| mt_de-en | gemma3-4b  | 1234 | +0.00 [+0.00, +0.00]       | -0.30 [-2.47, +2.17]         |
| mt_de-en | llama31-8b | 17   | -16.50 [-19.47, -13.64]    | -1.19 [-4.05, +1.88]         |
| mt_de-en | llama31-8b | 42   | -13.54 [-16.50, -11.07]    | -0.40 [-3.26, +2.57]         |
| mt_de-en | llama31-8b | 1234 | -23.32 [-26.78, -19.96]    | -1.28 [-4.35, +1.58]         |
| mt_de-en | llama32-3b | 17   | +0.00 [+0.00, +0.00]       | +0.89 [-1.38, +3.26]         |
| mt_de-en | llama32-3b | 42   | +0.00 [+0.00, +0.00]       | +2.47 [+0.00, +5.04]         |
| mt_de-en | llama32-3b | 1234 | -0.10 [-0.40, +0.00]       | +0.10 [-2.57, +2.87]         |
| units    | gemma3-12b | 17   | +0.50 [-1.70, +2.80]       | +0.70 [-0.90, +2.30]         |
| units    | gemma3-12b | 42   | -1.40 [-2.70, -0.20]       | +0.40 [-1.10, +2.00]         |
| units    | gemma3-12b | 1234 | -4.90 [-7.00, -3.10]       | +0.50 [-0.90, +1.90]         |

Table 7: Completed exploratory reverse-generation contrasts (percentage points), with 99% paired cluster-bootstrap intervals. Five primary contrasts are adjusted within each cell; no adjustment across cells. All proof-valid original and scale/domain-extension cells are included; failed-gate cells remain eligibility boundaries rather than tuned comparisons.

pendence; shared numerical structure limits interpretation of the units/GSM8K comparison.

All systems are evaluated together in fresh campaigns with adapter-effectiveness checks. Report strict IFEval and strict GSM8K endpoints separately; loose instruction checks and flexible numeric extraction remain secondary. Pair endpoint changes within each campaign and show seed variation. A concentration claim is scoped to this declared panel; unrelated benchmark scales are not subtracted as if interchangeable.

**Scoped finding.** General retention depends on the endpoint and task (Table 12). Formatting/Gemma4B loses numerical reasoning under forward SFT while instruction-following changes are small; replay recovers more of that reasoning endpoint than mixed-direction SFT. Transla-

tion/Llama3B loses instruction following while improving numerical reasoning. Reverse-task recovery therefore does not establish general-ability preservation, and the evidence does not support a universal disproportionate-loss claim across benchmarks.

### B.3 Independent real-data transfer

Existing German-to-English checkpoints are evaluated on the independently sourced OPUS-100 German–English test split.<sup>1</sup> The panel includes Llama3B and Gemma4B with the registered seeds and the same SFT, replay, mixed-direction, contrastive, and additional-CE systems. Dataset revision, character-length inclusion rules, directions, overlap exclusions, and scorer are frozen before

<sup>1</sup><https://huggingface.co/datasets/Helsinki-NLP/opus-100>.

| Task     | Model      | Seed | CL-fwd vs extra CE      | CL-fwd vs shuffled      | CL-mix5 vs extra CE  |
|----------|------------|------|-------------------------|-------------------------|----------------------|
| fmt      | Gemma 4B   | 17   | +0.04 [-0.84, +0.96]    | -29.20 [-31.80, -26.72] | -0.08 [-0.72, +0.56] |
| fmt      | Gemma 4B   | 42   | -16.04 [-18.00, -14.20] | -19.44 [-21.52, -17.48] | -0.44 [-1.28, +0.44] |
| fmt      | Gemma 4B   | 1234 | +8.20 [+6.80, +9.64]    | -22.16 [-25.08, -19.24] | -0.60 [-1.24, +0.00] |
| fmt      | llama31-8b | 17   | +2.24 [+1.52, +3.04]    | +2.24 [+1.52, +3.04]    | +0.44 [-0.16, +1.00] |
| fmt      | llama31-8b | 42   | -21.52 [-23.64, -19.44] | -33.00 [-35.40, -30.52] | +0.16 [-0.32, +0.68] |
| fmt      | llama31-8b | 1234 | +0.00 [+0.00, +0.00]    | +0.00 [+0.00, +0.00]    | -0.48 [-1.24, +0.28] |
| fmt      | Llama 3B   | 17   | -0.48 [-0.96, -0.08]    | -0.68 [-1.16, -0.24]    | +1.76 [+0.56, +2.96] |
| fmt      | Llama 3B   | 42   | -1.76 [-2.48, -1.12]    | -3.40 [-4.32, -2.52]    | +0.64 [-0.44, +1.68] |
| fmt      | Llama 3B   | 1234 | -0.36 [-0.88, +0.12]    | +0.28 [+0.08, +0.60]    | +2.76 [+1.48, +4.00] |
| mt_de-en | gemma3-12b | 17   | +0.10 [+0.00, +0.40]    | -0.10 [-0.40, +0.00]    | +0.49 [-2.17, +2.96] |
| mt_de-en | gemma3-12b | 42   | +0.00 [+0.00, +0.00]    | -0.10 [-0.40, +0.00]    | -1.19 [-3.56, +1.19] |
| mt_de-en | gemma3-12b | 1234 | +0.00 [+0.00, +0.00]    | +0.00 [+0.00, +0.00]    | -0.40 [-2.87, +2.08] |
| mt_de-en | Gemma 4B   | 17   | +0.00 [+0.00, +0.00]    | +0.00 [+0.00, +0.00]    | -0.59 [-2.77, +1.58] |
| mt_de-en | Gemma 4B   | 42   | +0.00 [+0.00, +0.00]    | +0.00 [+0.00, +0.00]    | +0.00 [-2.27, +2.27] |
| mt_de-en | Gemma 4B   | 1234 | +0.00 [+0.00, +0.00]    | +0.00 [+0.00, +0.00]    | +1.09 [-1.09, +3.36] |
| mt_de-en | llama31-8b | 17   | -18.18 [-21.34, -15.12] | +0.00 [+0.00, +0.00]    | -1.88 [-5.04, +1.28] |
| mt_de-en | llama31-8b | 42   | -14.53 [-17.49, -11.96] | -0.20 [-0.59, +0.00]    | -0.40 [-3.26, +2.47] |
| mt_de-en | llama31-8b | 1234 | -21.74 [-25.10, -18.38] | -0.10 [-0.49, +0.00]    | -1.09 [-4.15, +1.68] |
| mt_de-en | Llama 3B   | 17   | +0.00 [+0.00, +0.00]    | -0.10 [-0.40, +0.00]    | +0.69 [-1.68, +3.16] |
| mt_de-en | Llama 3B   | 42   | +0.00 [+0.00, +0.00]    | -0.10 [-0.40, +0.00]    | +1.78 [-0.59, +4.25] |
| mt_de-en | Llama 3B   | 1234 | -0.10 [-0.40, +0.00]    | -0.10 [-0.40, +0.00]    | -0.69 [-3.26, +1.88] |
| units    | gemma3-12b | 17   | +11.60 [+8.50, +14.90]  | -3.80 [-5.60, -2.00]    | +0.50 [-1.00, +2.00] |
| units    | gemma3-12b | 42   | -1.10 [-2.50, +0.20]    | -0.10 [-1.80, +1.60]    | +1.00 [-0.50, +2.60] |
| units    | gemma3-12b | 1234 | -2.60 [-4.60, -0.80]    | -4.80 [-7.00, -2.70]    | +0.80 [-0.60, +2.30] |

Table 8: The remaining three registered primary reverse-generation contrasts, in percentage points with 99% paired cluster-bootstrap intervals. Together with Table 7, these display all five primary comparisons for every completed cell. Extra CE is an exposure/compute proxy, not matched measured FLOPs. Degenerate zero intervals at an observed generation floor do not establish equivalence. Results remain exploratory; new scale/domain extensions are pending.

| Registered evaluation | Model/task cells | Training seeds | Campaigns |
|-----------------------|------------------|----------------|-----------|
| General probes        | 3                | 17,42,1234     | 9         |
| Prompt/template tests | 4                | 17,42,1234     | 12        |
| Independent transfer  | 2                | 17,42,1234     | 6         |
| Recipe evaluations    | 2                | 17,42,1234     | 6         |

Table 9: Frozen paper-completion panel. Probe corpora retain 541 IFEval prompts and 1319 GSM8K questions; independent transfer retains 1688 OPUS translation pairs. Synthetic magnitude/template tests retain 1000 instances each. Recipe packs train 48 new adapters in isolated directories. Counts are planned workloads and audited corpus sizes, not completed results.

generations. A new prompt on an existing semantic corpus remains a diagnostic rather than independent transfer evidence.

Overlap auditing includes all local training, validation, replay, and original test sources. Gold, echo, empty, and invalid-output oracles run on compute. Original model-specific COMET thresholds are retained. Strict success, continuous COMET, chrF++, echo, and off-target rates remain separate; checkpoint choice, decoding, and scoring are not tuned on this corpus.

Generate and score the full fresh base first, using the resident engine retained for tuned systems. Failure of the unchanged eligibility conjunction remains boundary evidence and blocks tuned generation. Small smokes establish engineering behavior,

not production eligibility. Report paired forward/reverse changes only for eligible campaigns.

**Completed measurements.** Every registered production transfer campaign fails the unchanged echo-validity conjunction. The base-only boundary trials are retained in Table 13; no tuned checkpoint comparison is eligible. These measurements therefore do not establish preservation on this independent corpus, and the failed criterion is not evidence of adapter collapse.

**Versioned diagnostics.** Amendment51 registers an exploratory repair extension after inspection of failed cells. The original failed gates remain intact. Separate versions test bounded signed C++ literal syntax, explicit serialization/conversion/logic con-

| Task     | Model      | Seed | Arm    | Forward (%) | Reverse (%) |
|----------|------------|------|--------|-------------|-------------|
| mt_de-en | llama32-3b | 17   | base   | 73.7        | 79.2        |
| mt_de-en | llama32-3b | 17   | sft    | 67.1        | 0.0         |
| mt_de-en | llama32-3b | 17   | replay | 68.6        | 59.0        |
| mt_de-en | llama32-3b | 17   | mix5   | 67.4        | 82.0        |
| mt_de-en | llama32-3b | 17   | mix50  | 67.4        | 82.9        |
| units    | gemma3-12b | 17   | base   | 33.7        | 35.8        |
| units    | gemma3-12b | 17   | sft    | 97.4        | 70.0        |
| units    | gemma3-12b | 17   | replay | 94.0        | 56.6        |
| units    | gemma3-12b | 17   | mix50  | 94.9        | 90.9        |

Table 10: Representative seed-17 preservation rates. Each model/task row set comes from one complete campaign. Additional doses, seed variation, paired intervals and measured cost records are retained in the source-backed analysis. The units boundary case is included independently of effect sign.

tracts, and a WebNLG extractor supplied with a global training-only predicate schema. Translation transfer additionally excludes source copies in both directions, retaining the original thresholds. Fresh compute-local oracles and base gates precede any tuned comparison. Revised criteria and prompts define new cells; their results will not replace original failures or support cross-version paired contrasts. Failed oracles and genuine capability limits remain reported boundaries.

#### B.4 Prompt and training-recipe robustness

The prompt panel includes translation/Llama3B, format conversion/Gemma4B, units/Gemma12B, and Python-C++/Llama8B. Frozen variants retain the original primary instruction and add fixed task/contract prefixes. Both directions use the same semantic instances, decoding limits, and scorers. Keep every frozen prompt, separating echo/off-target output, semantic error, and measurement failure. Best-prompt summaries remain exploratory recovery diagnostics.

Synthetic extrapolation additionally uses exact unit magnitudes outside original support and behavior-disjoint piecewise-quadratic Python/C++ functions. Original allowlisted scorers and input ranges remain unchanged; gold outputs are audited on all generated instances on compute. These tests do not establish unrestricted natural-code, loop/list, or independent-domain generality.

The recipe panel fixes lower/higher learning-rate and LoRA-rank variants around the original recipe, varying the factors separately. Parameter values, cells, seeds, and primary contrasts are fixed in the recorded protocol. Preserve alpha/rank ratio, effective batch, paired-data identity, direction mixture, sequence limit, epoch count, and optimizer schedule. Audit realized exposure, step

count, and GPU cost. SFT and mixed-direction SFT are compared on format conversion/Llama3B and units/Gemma12B, retaining collapse and null cases.

Store new adapters separately. Regenerate the original arms, base, and all recipe variants together in each campaign. Retain every adverse/null outcome and report paired intervals and training-seed ranges; no best-recipe selection supports the primary claim.

**Completed measurements.** The frozen prompt panel retains directional loss under forward SFT for translation and format conversion; unit conversion remains a null-collapse case under the original recipe (Table 14). Preservation is not uniform across tasks or recipes: the Python-C++ mixture has large seed-dependent forward costs, and the higher learning-rate unit recipe includes reverse loss as well as improvement (Table 15). The unit boundary case is therefore scoped to the original recipe. Every frozen prompt and recipe is retained; selecting the best variant does not support the primary claim. These seed ranges are descriptive, and endpoint scales are not pooled into an overall preservation score.

#### B.5 Contribution and reproducibility

The final abstract and conclusion should state the supported contribution relative to prior directional degradation and reverse-training work: the preservation frontier, direction and compute controls for objective attribution, and limits revealed by gates and null cells. Generality, disproportionate loss, inexpensive preservation, and recovery are separate claims with separate evidence requirements. A favorable contrastive result supports a scoped objective contribution; a null or generation-floor result does not establish equivalence.

| Task     | Model      | Arm           | Minutes | Steps | CE corpus tokens |
|----------|------------|---------------|---------|-------|------------------|
| mt_de-en | llama32-3b | sft           | 8.9     | 306   | 1096262          |
| mt_de-en | llama32-3b | mix5          | 8.9     | 306   | 1096262          |
| mt_de-en | llama32-3b | mix50         | 9.0     | 306   | 1096262          |
| mt_de-en | llama32-3b | replay        | 8.9     | 306   | 1096110          |
| mt_de-en | llama32-3b | sft           | 8.9     | 306   | 1096262          |
| mt_de-en | llama32-3b | mix5          | 8.9     | 306   | 1096262          |
| mt_de-en | llama32-3b | replay        | 8.9     | 306   | 1096110          |
| mt_de-en | llama32-3b | cl_fwd        | 241.9   | 306   | 1096262          |
| mt_de-en | llama32-3b | cl_mix5       | 62.2    | 306   | 1096262          |
| mt_de-en | llama32-3b | sft_extra_ce  | 123.2   | 306   | 1096262          |
| mt_de-en | llama32-3b | mix5_extra_ce | 119.1   | 306   | 1096262          |
| fnt      | gemma3-4b  | sft           | 30.6    | 306   | 1927990          |
| fnt      | gemma3-4b  | mix5          | 30.7    | 306   | 1927882          |
| fnt      | gemma3-4b  | replay        | 30.2    | 306   | 1927978          |
| fnt      | gemma3-4b  | cl_fwd        | 194.3   | 306   | 1927990          |
| fnt      | gemma3-4b  | cl_mix5       | 193.5   | 306   | 1927882          |
| fnt      | gemma3-4b  | sft_extra_ce  | 191.3   | 306   | 1927990          |
| fnt      | gemma3-4b  | mix5_extra_ce | 194.0   | 306   | 1927882          |
| units    | gemma3-12b | sft           | 21.6    | 306   | 788819           |
| units    | gemma3-12b | mix50         | 22.1    | 306   | 788819           |
| units    | gemma3-12b | replay        | 21.5    | 306   | 799547           |
| units    | gemma3-12b | sft           | 21.6    | 306   | 788819           |
| units    | gemma3-12b | mix5          | 15.6    | 306   | 788819           |
| units    | gemma3-12b | replay        | 21.5    | 306   | 799547           |
| units    | gemma3-12b | cl_fwd        | 82.8    | 306   | 788819           |
| units    | gemma3-12b | cl_mix5       | 83.0    | 306   | 788819           |
| units    | gemma3-12b | sft_extra_ce  | 166.9   | 306   | 788819           |
| units    | gemma3-12b | mix5_extra_ce | 168.0   | 306   | 788819           |

Table 11: Recorded seed-17 trainer runtimes (minutes), optimizer steps, and sequence tokens in the tokenized CE corpus (before epoch repetition). Auxiliary encoded-side exposure remains separately recorded in the analysis; these token counts are not total objective FLOPs. Runtime varies with GPU/training layout and is not a claim of matched wall time. Missing measurements are unavailable, never inferred.

**Supported contribution.** The study tests preservation of an existing reverse capability using ordinary supervised training with direction and exposure controls. Its contribution is the audited comparison of both task directions, objective attribution against matched baselines, and explicit limits from failed gates, null cells and sensitivity to training recipes. Exploratory recovery diagnostics qualify hypotheses about accessibility; they do not establish a universal internal mechanism.

The local reproducibility package includes permitted corpora or versioned reconstruction recipes; overlap and scorer audits; manifests and environment settings; irreproducible per-instance generations and their hashes; analysis entrypoints; and figure/table provenance. Failed gates, withdrawals, and corrected replacements remain traceable. Dataset/weight redistribution follows their licenses.

Before release, replay analysis and the PDF build from a clean environment, check reported values against generating trials and references, and remove identifying local names/paths while retaining a hash mapping for transformed artifacts.

Missing historical code versions must be recorded rather than implying exact training reproducibility. This is a local artifact deliverable; no release or submission has been performed.

## C Budget-limited explanatory diagnostics

Amendment54 limits new compute to five fixed seed-17 diagnostics and required engineering smokes. The original 24-job panel remains registered; unrun jobs are deferred. Selection uses the time budget and distinct scientific questions, not new diagnostic outcomes. Completed earlier seeds are retained. These single-seed analyses are exploratory; paired intervals measure test-pair uncertainty, not training-seed stability. All candidates, controls, bands and local steps remain in the evidence bundle. Proof-valid diagnostic jobs at this snapshot: 5 of 5. In the completed Llama formatting and translation diagnostics, forward SFT lowers gold completion loss in the forward direction and raises it in reverse; mixed-direction SFT lowers reverse loss relative to SFT with small mean

| Cell             | Endpoint | Seeds | SFT-base       | mix5-SFT      | Replay-SFT     |
|------------------|----------|-------|----------------|---------------|----------------|
| fmt/Gemma4B      | gsm8k    | 3     | [-56.6, -36.3] | [+0.5, +21.8] | [+24.7, +46.0] |
| fmt/Gemma4B      | ifeval   | 3     | [-1.3, +0.6]   | [-3.0, +0.9]  | [-8.3, -4.1]   |
| mt_de-en/Llama3B | gsm8k    | 3     | [+6.6, +8.7]   | [-2.7, +1.4]  | [-13.7, +10.6] |
| mt_de-en/Llama3B | ifeval   | 3     | [-10.9, -9.6]  | [-0.4, +4.4]  | [+0.2, +3.0]   |
| units/Gemma12B   | gsm8k    | 3     | [+1.2, +2.0]   | [-0.7, +0.9]  | [-10.4, -9.1]  |
| units/Gemma12B   | ifeval   | 3     | [-3.0, +1.5]   | [+0.0, +3.5]  | [-13.5, -3.0]  |

Table 12: Strict general-ability endpoint changes: descriptive minimum/maximum across all completed training seeds, in percentage points. Each difference uses its own fresh campaign baseline. These ranges are not confidence intervals. Per-seed paired intervals, all objective controls and overlap exclusions remain in the evidence artifact; endpoints are not pooled.

| Model      | Seed | Direction | Base (%) | Format failure (%) | Echo (%) | Campaign |
|------------|------|-----------|----------|--------------------|----------|----------|
| gemma3-4b  | 1234 | forward   | 33.9     | 10.1               | 6.6      | blocked  |
| gemma3-4b  | 1234 | reverse   | 44.9     | 6.3                | 0.0      | blocked  |
| gemma3-4b  | 17   | forward   | 33.9     | 10.0               | 6.6      | blocked  |
| gemma3-4b  | 17   | reverse   | 44.8     | 6.3                | 0.0      | blocked  |
| gemma3-4b  | 42   | forward   | 33.9     | 10.1               | 6.6      | blocked  |
| gemma3-4b  | 42   | reverse   | 44.8     | 6.3                | 0.0      | blocked  |
| llama32-3b | 1234 | forward   | 33.7     | 10.6               | 6.8      | blocked  |
| llama32-3b | 1234 | reverse   | 53.0     | 6.8                | 0.0      | blocked  |
| llama32-3b | 42   | forward   | 33.7     | 10.6               | 6.8      | blocked  |
| llama32-3b | 42   | reverse   | 52.9     | 6.8                | 0.0      | blocked  |
| llama32-3b | 17   | forward   | 33.7     | 10.6               | 6.8      | blocked  |
| llama32-3b | 17   | reverse   | 52.8     | 6.8                | 0.0      | blocked  |

Table 13: Completed OPUS transfer eligibility measurements with unchanged thresholds. Each campaign requires both directions to satisfy the frozen rate, format-failure, and echo conjunction. Failed campaigns preserve base-only trials and do not generate tuned comparisons; an echo failure is a criterion-validity boundary, not evidence of adapter collapse.

forward changes (Table 17). The local SFT gradient cosines are near zero (Table 18); these checkpoints provide little support for a simple persistent gradient-conflict account. Gradients at these final checkpoints do not reconstruct interference along the training trajectory. This descriptive evidence concerns answer likelihood and does not establish stored knowledge. The Gemma translation diagnostic differs: reverse gold completion loss improves relative to base despite collapsed free generation. Absolute answer loss therefore does not uniformly track collapse. In the three collapse cells, per-token gold-over-copy preference falls under forward SFT and recovers under mixed-direction SFT; the units null cell retains strong preference (Table 19). Total and per-token rankings are reported together because candidate lengths differ. Translation has multiple valid answers; a restricted candidate ranking is neither correctness nor a measure of retained knowledge. At the SFT checkpoint, both tested reverse and mixed local-step sizes lower reverse held-out loss in every cell, including the null cell; this is not a collapse-specific signature or a demonstrated gen-

eration repair. All checkpoint states, update directions, step sizes and both endpoints are retained in the bundle. Formatting middle-band removals improve reverse success beyond all matched controls with small forward changes, whereas the earliest band harms forward performance without consistent reverse recovery (Table 20). This supports depth selectivity in one cell, with unadjusted intervals across multiple interventions, rather than a universal causal mechanism.

| Cell         | Setting                           | Seeds | $\Delta F_{S-B}$ | $\Delta R_{S-B}$ | $\Delta F_{M-S}$ | $\Delta R_{M-S}$ |
|--------------|-----------------------------------|-------|------------------|------------------|------------------|------------------|
| fmt/G4B      | original/contract_first           | 3     | [+20.0, +20.2]   | [-58.5, -27.8]   | [+0.0, +0.1]     | [+51.9, +82.1]   |
| fmt/G4B      | original/primary                  | 3     | [+21.4, +21.7]   | [-60.0, -27.9]   | [+0.0, +0.2]     | [+51.8, +83.8]   |
| fmt/G4B      | original/reword                   | 3     | [+22.2, +22.2]   | [-59.8, -27.8]   | [-0.0, +0.1]     | [+51.8, +83.5]   |
| mt_de-en/L3B | original/contract_first           | 3     | [-4.4, -4.2]     | [-79.0, -78.7]   | [-0.5, +0.6]     | [+79.9, +81.1]   |
| mt_de-en/L3B | original/primary                  | 3     | [-7.7, -5.5]     | [-79.7, -79.6]   | [-0.3, +0.7]     | [+79.2, +82.0]   |
| mt_de-en/L3B | original/reword                   | 3     | [-4.6, -3.1]     | [-79.1, -78.9]   | [-0.7, +0.5]     | [+78.8, +81.6]   |
| py_cpp/L8B   | original/contract_first           | 3     | [+4.4, +4.4]     | [-100.0, -100.0] | [-92.5, +0.0]    | [+0.0, +50.0]    |
| py_cpp/L8B   | original/primary                  | 3     | [+15.8, +15.8]   | [-100.0, -100.0] | [-97.8, +0.0]    | [+0.0, +48.7]    |
| py_cpp/L8B   | original/reword                   | 3     | [+6.4, +6.4]     | [-100.0, -100.0] | [-89.2, +0.0]    | [+0.0, +49.9]    |
| py_cpp/L8B   | py_cpp_pieewise_quadratic/primary | 3     | [-68.2, -68.2]   | [+0.0, +0.0]     | [+0.0, +0.0]     | [+99.3, +100.0]  |
| units/G12B   | original/contract_first           | 3     | [+63.6, +63.6]   | [+32.5, +36.6]   | [-2.3, -1.4]     | [+17.1, +20.7]   |
| units/G12B   | original/primary                  | 3     | [+63.3, +63.3]   | [+34.0, +38.0]   | [-2.1, -1.4]     | [+17.4, +21.3]   |
| units/G12B   | original/reword                   | 3     | [+65.3, +65.6]   | [+47.3, +50.7]   | [-2.3, -1.6]     | [+17.4, +20.6]   |
| units/G12B   | units_magnitude/primary           | 3     | [+35.1, +47.2]   | [+36.7, +40.0]   | [-1.4, +5.0]     | [+6.0, +8.5]     |

Table 14: Completed robust panel: ranges across training seeds of within-campaign strict-success changes (percentage points).  $S$ ,  $B$ , and  $M$  denote forward SFT, same-pass base, and the registered directional mixture. Recipe rows compare matched SFT/mixture variants. Mixture doses are 5% for translation/format and 50% for units/Python-C++; L/G identify Llama/Gemma and B denotes billions of parameters. Ranges are descriptive seed variation, not confidence intervals; per-campaign paired intervals and every control remain in the analysis artifact.

| Cell       | Setting   | Seeds | $\Delta F_{S-B}$ | $\Delta R_{S-B}$ | $\Delta F_{M-S}$ | $\Delta R_{M-S}$ |
|------------|-----------|-------|------------------|------------------|------------------|------------------|
| fmt/L3B    | lr_high   | 3     | [+39.5, +39.5]   | [-51.1, -49.2]   | [-0.2, -0.0]     | [+95.4, +98.2]   |
| fmt/L3B    | lr_low    | 3     | [+38.7, +39.1]   | [-43.0, -40.6]   | [+0.2, +0.6]     | [+82.6, +86.6]   |
| fmt/L3B    | original  | 3     | [+39.4, +39.5]   | [-50.8, -49.8]   | [-0.1, +0.2]     | [+92.2, +95.7]   |
| fmt/L3B    | rank_high | 3     | [+39.0, +39.3]   | [-51.1, -51.0]   | [-0.0, +0.4]     | [+95.8, +96.6]   |
| fmt/L3B    | rank_low  | 3     | [+38.2, +39.4]   | [-50.4, -45.4]   | [-0.0, +1.1]     | [+90.8, +92.9]   |
| units/G12B | lr_high   | 3     | [+63.6, +63.7]   | [-28.2, +12.1]   | [-4.2, -1.1]     | [+43.7, +84.6]   |
| units/G12B | lr_low    | 3     | [+62.8, +63.1]   | [+36.1, +37.4]   | [-3.4, -2.5]     | [+14.1, +16.2]   |
| units/G12B | original  | 3     | [+63.3, +63.6]   | [+34.1, +37.8]   | [-2.3, -1.2]     | [+17.7, +21.2]   |
| units/G12B | rank_high | 3     | [+63.4, +64.3]   | [+13.7, +27.9]   | [-2.7, -1.9]     | [+27.7, +41.1]   |
| units/G12B | rank_low  | 3     | [+62.8, +63.6]   | [+25.6, +37.6]   | [-3.1, -0.4]     | [+15.9, +28.3]   |

Table 15: Completed recipe panel: ranges across training seeds of within-campaign strict-success changes (percentage points).  $S$ ,  $B$ , and  $M$  denote forward SFT, same-pass base, and the registered directional mixture. Recipe rows compare matched SFT/mixture variants. Mixture doses are 5% for translation/format and 50% for units/Python-C++; L/G identify Llama/Gemma and B denotes billions of parameters. Ranges are descriptive seed variation, not confidence intervals; per-campaign paired intervals and every control remain in the analysis artifact.

| Deliverable              | Required evidence                                                          | Current status                                                  |
|--------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------|
| Preservation figure      | Both directions, intervals, seeds, doses, exposure and measured cost       | Audited synthesis integrated; interpretation pending            |
| General-ability table    | Frozen panel, overlap exclusions, same-pass changes and intervals          | All frozen campaigns complete; interpretation pending           |
| Independent transfer     | Corpus/version, eligibility, both directions, failures and hashes          | All frozen OPUS gates fail; base-only boundaries retained       |
| Robustness table         | Frozen prompts/recipes, both directions, costs and all outcomes            | All prompt/recipe campaigns complete; scoped results integrated |
| Reproducibility manifest | Splits/scorers, environment, manifests, trial hashes and analysis commands | Local provenance retained; release audit pending                |

Table 16: Reporting checklist. Registration, auditing, and queue admission are separate from completed experimental evidence.

| Task     | Model      | Direction | SFT minus base          | mix5 minus SFT          |
|----------|------------|-----------|-------------------------|-------------------------|
| fmt      | llama32-3b | forward   | -0.117 [-0.132, -0.101] | +0.000 [-0.000, +0.001] |
| fmt      | llama32-3b | reverse   | +0.121 [+0.104, +0.139] | -0.288 [-0.331, -0.247] |
| mt_de-en | llama32-3b | forward   | -0.277 [-0.336, -0.220] | +0.004 [-0.003, +0.012] |
| mt_de-en | llama32-3b | reverse   | +0.103 [+0.075, +0.131] | -0.181 [-0.206, -0.155] |
| mt_de-en | gemma3-4b  | forward   | -2.352 [-2.547, -2.168] | -0.002 [-0.012, +0.008] |
| mt_de-en | gemma3-4b  | reverse   | -1.008 [-1.122, -0.893] | -0.419 [-0.459, -0.378] |
| units    | gemma3-12b | forward   | -2.587 [-2.829, -2.358] | +0.016 [-0.005, +0.053] |
| units    | gemma3-12b | reverse   | -2.612 [-2.832, -2.405] | -0.381 [-0.503, -0.273] |

Table 17: Gold completion NLL per token, paired differences with descriptive 95% bootstrap intervals on 128 fixed pairs. Negative values favor the first system. Candidate likelihood does not establish generation correctness or stored knowledge. All seeds here are 17.

| Task     | Model      | SFT    | Replay | mix5   |
|----------|------------|--------|--------|--------|
| fmt      | llama32-3b | -0.018 | +0.004 | +0.010 |
| mt_de-en | llama32-3b | -0.000 | +0.048 | +0.059 |
| mt_de-en | gemma3-4b  | +0.021 | +0.012 | +0.031 |
| units    | gemma3-12b | -0.013 | +0.009 | -0.026 |

Table 18: Forward/reverse validation-gradient cosine on 16 pairs in local LoRA A/B coordinates. Gemma excludes inactive vision factors. These coordinate-dependent quantities do not by themselves establish interference or causality.

| Task     | Model      | Direction | Base          | SFT           | mix5          |
|----------|------------|-----------|---------------|---------------|---------------|
| fmt      | llama32-3b | forward   | 0.695 / 0.719 | 1.000 / 1.000 | 1.000 / 1.000 |
| fmt      | llama32-3b | reverse   | 0.750 / 0.672 | 0.000 / 0.008 | 1.000 / 1.000 |
| mt_de-en | llama32-3b | forward   | 0.344 / 0.195 | 0.484 / 0.234 | 0.445 / 0.242 |
| mt_de-en | llama32-3b | reverse   | 0.062 / 0.180 | 0.000 / 0.000 | 0.320 / 0.562 |
| mt_de-en | gemma3-4b  | forward   | 0.375 / 0.289 | 0.461 / 0.398 | 0.359 / 0.273 |
| mt_de-en | gemma3-4b  | reverse   | 0.406 / 0.586 | 0.000 / 0.000 | 0.430 / 0.562 |
| units    | gemma3-12b | forward   | 0.828 / 0.875 | 1.000 / 1.000 | 0.992 / 1.000 |
| units    | gemma3-12b | reverse   | 0.586 / 0.547 | 1.000 / 0.977 | 1.000 / 1.000 |

Table 19: Fraction of 128 fixed pairs where gold completion NLL is strictly lower than copied-input NLL: total / per-token ranking. Candidate lengths differ, so these rankings need not agree. Both directions and the same focal systems as the likelihood contrasts are shown; all eight systems and shuffled-target candidates remain in the bundle. Single training seed 17.

| Removed band | Matched control | Direction | Success difference      |
|--------------|-----------------|-----------|-------------------------|
| band0_remove | band0_scale     | forward   | -0.043 [-0.062, -0.025] |
| band0_remove | band0_scale     | reverse   | -0.006 [-0.014, +0.000] |
| band0_remove | band0_random101 | forward   | -0.039 [-0.057, -0.021] |
| band0_remove | band0_random101 | reverse   | -0.014 [-0.023, -0.004] |
| band0_remove | band0_random202 | forward   | -0.037 [-0.055, -0.020] |
| band0_remove | band0_random202 | reverse   | +0.000 [+0.000, +0.000] |
| band1_remove | band1_scale     | forward   | -0.010 [-0.021, +0.000] |
| band1_remove | band1_scale     | reverse   | +0.096 [+0.070, +0.121] |
| band1_remove | band1_random101 | forward   | -0.008 [-0.018, +0.000] |
| band1_remove | band1_random101 | reverse   | +0.094 [+0.068, +0.119] |
| band1_remove | band1_random202 | forward   | -0.008 [-0.018, +0.000] |
| band1_remove | band1_random202 | reverse   | +0.100 [+0.074, +0.125] |
| band2_remove | band2_scale     | forward   | -0.002 [-0.012, +0.006] |
| band2_remove | band2_scale     | reverse   | +0.084 [+0.061, +0.107] |
| band2_remove | band2_random101 | forward   | +0.000 [-0.010, +0.010] |
| band2_remove | band2_random101 | reverse   | +0.082 [+0.057, +0.107] |
| band2_remove | band2_random202 | forward   | +0.002 [-0.008, +0.012] |
| band2_remove | band2_random202 | reverse   | +0.088 [+0.062, +0.113] |
| band3_remove | band3_scale     | forward   | +0.002 [+0.000, +0.006] |
| band3_remove | band3_scale     | reverse   | +0.037 [+0.021, +0.055] |
| band3_remove | band3_random101 | forward   | +0.006 [+0.000, +0.014] |
| band3_remove | band3_random101 | reverse   | +0.033 [+0.018, +0.051] |
| band3_remove | band3_random202 | forward   | +0.008 [+0.002, +0.016] |
| band3_remove | band3_random202 | reverse   | +0.043 [+0.025, +0.061] |

Table 20: All fixed-depth removals versus retained-norm-matched controls for formatting/Llama3B (seed17), strict no-echo success differences on 512 pairs with descriptive 95% paired intervals. Retained and edit norms are both recorded; only retained norms are matched. No best band or control is selected.